SOFTWARE REQUIREMENTS SPECIFICATION

for

INVESTCIRCLE

Version 1.0

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1 Introduction

1.1 Purpose

Retail investors research stocks across social media, discussion forums, financial news websites and informal stock-tip communities. On these platforms predictions are rarely recorded in a structured form, their outcomes are not tracked, and credibility is judged by followers or likes rather than by historical accuracy. So, "InvestCircle" is the solution. "InvestCircle" is a web-based, peer-to-peer financial research platform that combines identity-based stock discussion, structured stock calls that are automatically evaluated against real market data, and a personalised scrollable market-news feed. The main concept of "InvestCircle" is to build a persistent, public track record for every structured call, so that credibility is earned rather than claimed.
This document is the Software Requirements Specification (SRS) of "InvestCircle" Version 1.0. It is derived from, and consistent with, the approved project proposal (UCS503). Requirements are identified as FR (functional), NFR (non-functional) and priorities follow MoSCoW: **Must** (MVP), **Should** (subsequent iterations), **Could** (optional extension).

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, project managers, users, testers and the course instructor. Further the discussion will provide all the internal, external, functional and also non-functional informations about "InvestCircle". Every requirement is written to be verifiable.
Key terms used in this document:

- **Call (stock call)** - a structured prediction made of ticker, direction (bullish/bearish), target price and time horizon, attached to a post.
- **Pending / Hit / Miss** - status of a call: not yet resolved / target reached within the horizon / target not reached.
- **Track record** - public history and accuracy score of a user's confirmed calls.
- **Watchlist** - a user-maintained list of tickers.
- **News Reel** - the scrollable, reels-style market-news feed.
- **Personal page** - a user's profile with posts, followers and track record.
- **MVP** - minimum viable product (initial deliverable).

1.3 Project Scope

"InvestCircle" creates a space for Readers, Contributors and Moderators to do financial research and discussion under real identities. It is for **research and discussion, not trading**: it will not execute trades, hold funds or connect to any brokerage account.
Objectives.

1. Accountability: keep a permanent public record of every structured call and its outcome, tied to the author's real identity.
2. Measurable credibility: use historical accuracy instead of likes or follower counts as the main credibility signal.
3. Consolidated research: bring discussion, news and watchlists into one platform.
4. Useful scrolling: offer a fast scrollable news feed that replaces unproductive short-form scrolling.
5. Reliability: resolve at least 95% of due calls automatically.
6. Engineering quality: a modular, secure, scalable system with at least 70% automated test coverage of backend and business logic.

In scope. User registration, authentication, profiles and follow/unfollow; a Discussions tab with posts, voting, comments and threaded replies; structured stock calls with automatic resolution and public accuracy record; watchlists; a News Reel personalised by watchlist; notifications; search and discovery; contributor statistics; basic moderation.
Optional extensions (outside the core MVP). Sector/sentiment tagging, AI-inferred call detection (always labelled "AI-inferred"), short video reactions to news ("Share Your Opinion"), advanced recommendations, analytics and rankings.
Out of scope. Trade execution, brokerage integration, payment processing, investment advice, novel price-prediction algorithms and native mobile apps (a responsive web interface is provided).
Stakeholders.

Stakeholder	Role	Main interest
Readers (retail investors)	Browse discussions, follow users, use News Reel and watchlists	Trustworthy research and relevant news
Contributors	Publish posts and structured calls	Earn credibility through a verifiable record
Moderators / Admins	Review reports, remove content, manage accounts	Platform quality and safety
Development team	Design, build, test, deploy	Clear, testable requirements
Course instructor	Reviews and evaluates the project	Consistency with the approved proposal
Market-data provider	Supplies prices and news (Alpha Vantage, Finnhub or similar)	Fair, rate-limited API use

Table 1.1: Stakeholders of InvestCircle

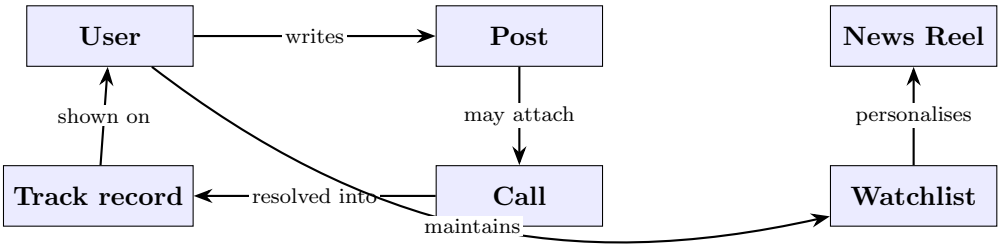


Figure 1.1: Entire work-flow

Figure 1.1 (Entire work-flow) is the overview of the project. A "User" writes a "Post"; a "Post" may attach a "Call"; every "Call" is resolved into the user's "Track record", which is shown on the user's profile. A user's "Watchlist" personalises the "News Reel". So, every entity is very interactive with each other.

2 Overall Description

2.1 Product Perspective

"InvestCircle" is a new, self-contained web application. It replaces the scattered, untracked way retail investors currently share stock opinions across social media and forums, and depends on external services only for market data. Main goal of this project is to make financial discussion accountable, transparent and measurable. It follows a layered client–server architecture (Figure 2.1):

- **Frontend:** React single-page application, responsive, with light and dark mode.
- **Backend API:** Node.js REST services for authentication, users, posts, comments, calls, watchlists and notifications.
- **Business logic:** call resolution, feed ranking, follower relationships, notification generation.
- **Database:** PostgreSQL. **Cache/queue:** Redis.
- **External services:** financial-market APIs and optional AI services behind a provider-abstraction layer.

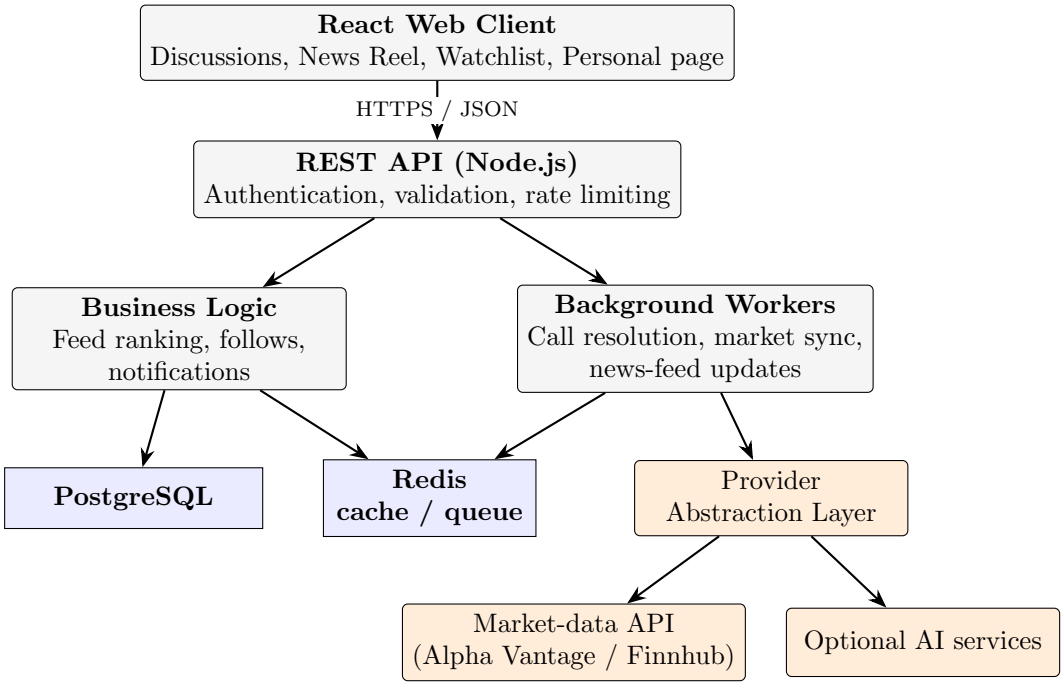


Figure 2.1: High-level system architecture

2.2 User Classes and Characteristics

"InvestCircle" has basically 4 types of users.

- Guest (unregistered)
- Registered User
  - Reader
  - Contributor (a user who has published structured calls)
- Moderator / Admin
- External systems (market-data provider, OAuth provider, optional AI service)

A Guest may view limited public content only and cannot post, vote, follow or comment. A Registered User posts, votes, comments, follows, manages watchlists, uses the News Reel and owns a Personal page. A Moderator / Admin reviews reported content, removes violating content and suspends accounts. Users need only basic web-browsing skill; moderators need medium technical skill.

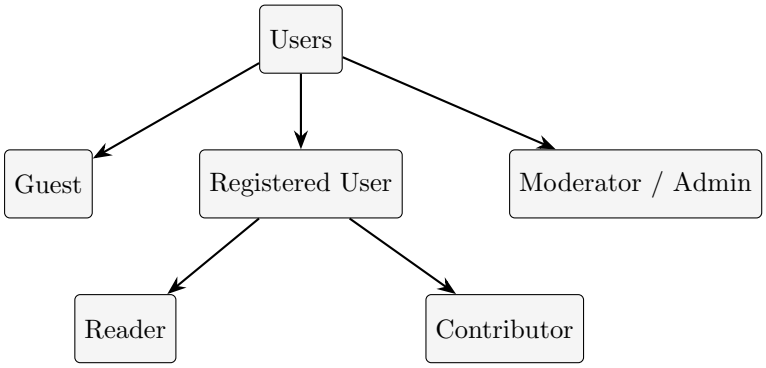


Figure 2.2: type of users

2.3 Product Functions

"InvestCircle" stores all posts, calls and call outcomes of its users and maintains their public track records. At a high level it shall allow users to register and keep a persistent profile; publish research posts with an optional structured call; vote, comment and reply in threads; follow other users and receive notifications; maintain watchlists; scroll a personalised News Reel; search users, stocks and discussions; and have every call automatically resolved as Hit or Miss.
Before using the main functions, users have to be registered. The principal data entities (Figure 2.3) are:

- **users** - user_id, full_name, email (unique), password_hash, oauth_provider, bio, role, verified, status, created_at.
- **posts** - post_id, author_id (foreign key), title, body, ticker, upvotes, downvotes, created_at, updated_at, is_deleted.
- **calls** - call_id, post_id (foreign key, unique), author_id, ticker, direction, target_price, reference_price, horizon_date, status (Pending/Hit/Miss), is_ai_inferred, resolved_at, resolution_price, data_source. A post can optionally reference one call; the flag separates explicit from AI-inferred calls.
- **comments** - comment_id, post_id, author_id, parent_comment_id (self-reference for threaded replies), body, created_at.
- **votes** - vote_id, post_id, user_id, value (+1/-1); one vote per user per post.
- **follows** - follower_id, followee_id, created_at.
- **watchlists** and **watchlist_items** - owner_id, name, ticker, added_at.
- **notifications** - recipient_id, type (follow/comment/call_resolved), reference_id, is_read, created_at.

- **news_items** - headline, source, published_at, ticker, category, url. **reactions** - news_id, user_id, video_ref (optional feature).

Foreign keys maintain referential integrity; indexes are applied to user IDs, tickers, call status/horizon and timestamps. Call is the main feature of all, so resolved calls are retained permanently and are immutable except for logged administrator corrections.

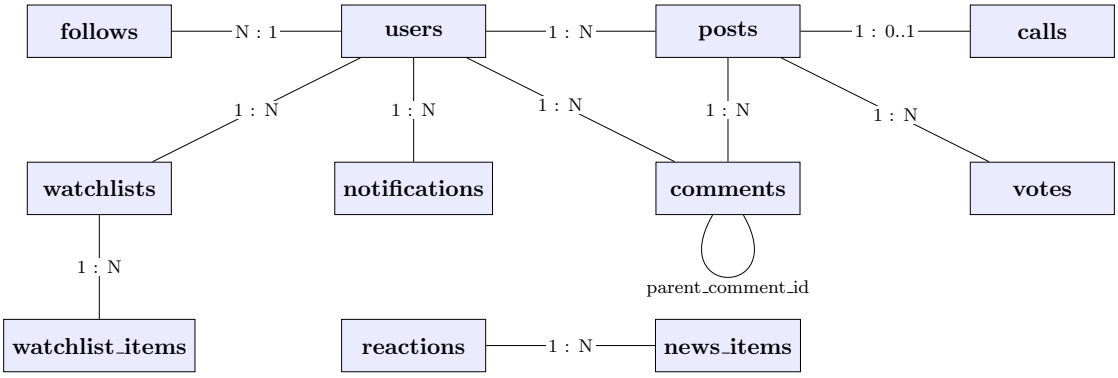


Figure 2.3: Data Flow Diagram (entity relationships)

2.4 Operating Environment

The website will be operate in any modern web browser (current Chrome, Firefox, Edge, Safari) on Windows, Mac, Linux, Android or iOS with a responsive layout. The server side runs as containerised Node.js services on common cloud hosting with managed PostgreSQL and Redis. An Internet connection and HTTPS are required.
External interfaces.

- **User interface:** responsive React UI with light and dark mode; three main sections (Discussions, News Reel, Personal page) in a persistent navigation bar; paginated feeds; an optional structured-call form with inline validation; a visible "AI-inferred" label on inferred calls; clear loading, empty and error states.
- **Hardware:** no special hardware (camera/microphone only for the optional video reactions).
- **Software:** market-data API (Alpha Vantage/Finnhub or similar, via a provider-abstraction layer, cached in Redis); optional OAuth provider; PostgreSQL through an ORM; Redis for cache and job queue; optional AI service (non-critical).
- **Communications:** HTTPS with JSON over REST; JWT in the Authorization header; cursor/page-based pagination.

Design and implementation constraints.

- Stack: React, Node.js REST backend, PostgreSQL, Redis, JWT authentication.
- Market data only through the external provider, subject to its rate limits and terms; provider isolated behind a service layer.
- Passwords hashed with bcrypt or Argon2; database access only through an ORM or parameterised queries.
- No trade execution or brokerage integration.
- Authentication, discussions, watchlists and explicit call tracking must stay fully functional if optional AI components are unavailable.
- Four-person team, four months, short iterations with explicit module ownership; Git and CI/CD.

Assumptions and dependencies.

- The provider supplies price history sufficient to evaluate calls, and free-tier limits are adequate with caching.
- Users register with a valid email address or an OAuth account.
- **Hit/Miss rule:** a bullish call is a Hit if the price reaches or exceeds the target at any point within the horizon; a bearish call is a Hit if it reaches or falls below the target (evaluated on daily high/low data); otherwise it is a Miss. This fixes the proposal's "checks whether the target price was reached" so that it can be tested unambiguously.

2.5 Design

Contributor activities have 7 steps - the core workflow of "InvestCircle":

1. A registered user creates a discussion post.
2. The user may attach a structured call (ticker, bullish/bearish, target price, time horizon).
3. The system stores the call with status Pending.
4. When the horizon expires, a background process obtains the relevant market price.
5. The system checks whether the target price was reached.
6. The call is marked Hit or Miss, and the resolution timestamp and data source are stored.
7. The result becomes part of the user's public track record on the Personal page.

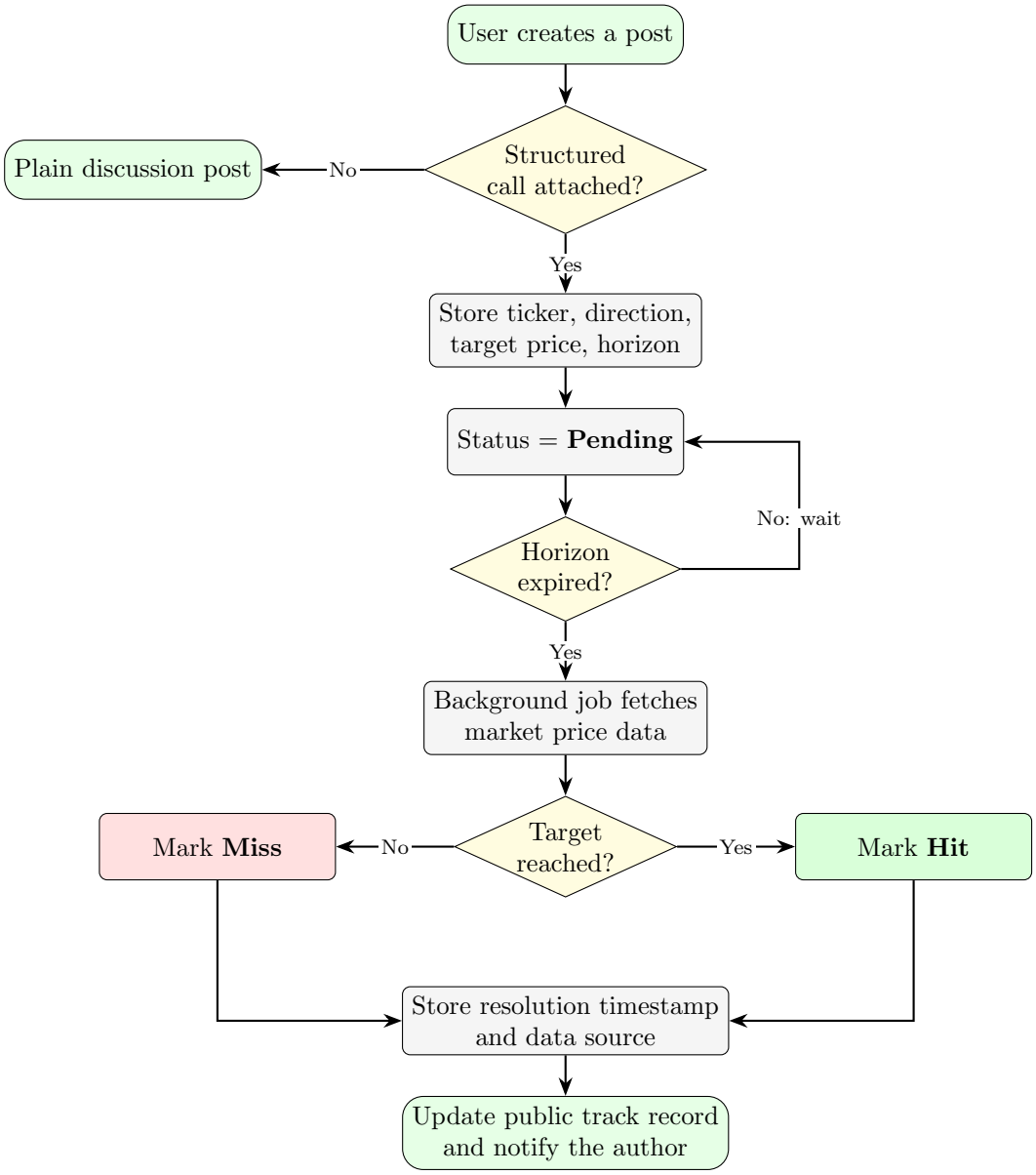


Figure 2.4: Stock-call resolution workflow

Reader activities have 3 steps - Register / log in, Browse (Discussions feed, News Reel, search), and Participate (vote, comment, reply, follow, maintain watchlists). Moderator / Admin activities are: review reported content, remove content, suspend accounts, and correct a resolved call (logged). Every user's Personal page contains identity information, biography, followers, following, own posts and the historical call results with an overall accuracy score.

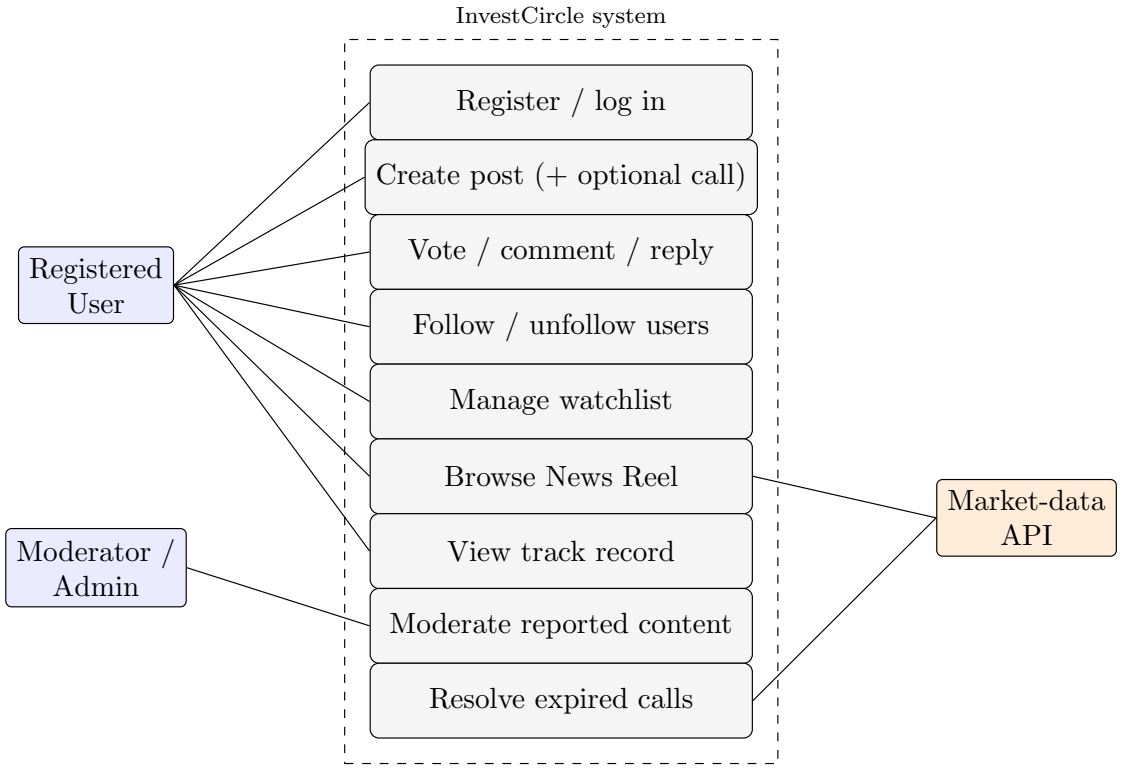


Figure 2.5: Use-case diagram (User, Moderator and Staff activities)

UC-1 Publish a structured call. Actor: registered user, who must be logged in. The user opens the composer, writes the research text, enables "Attach call" and enters direction, target price and horizon. The system validates the input (positive target price, known ticker, future horizon), stores the post and call as Pending and shows it in the feed with a Pending badge. If validation fails an error is shown and nothing is stored.
UC-2 Resolve an expired call. Actor: background worker and market-data API. The scheduler selects due Pending calls, fetches price data (cache first), applies the Hit/Miss rule, stores status, timestamp and data source, recomputes the author's accuracy and creates a notification. If the provider is unavailable, the call stays Pending and is retried with back-off.

Supplementary Diagrams (Chapter 2)

In the DFDs, ellipses are processes, thick rectangles are external entities and the plain rectangles D1–D6 are data stores.

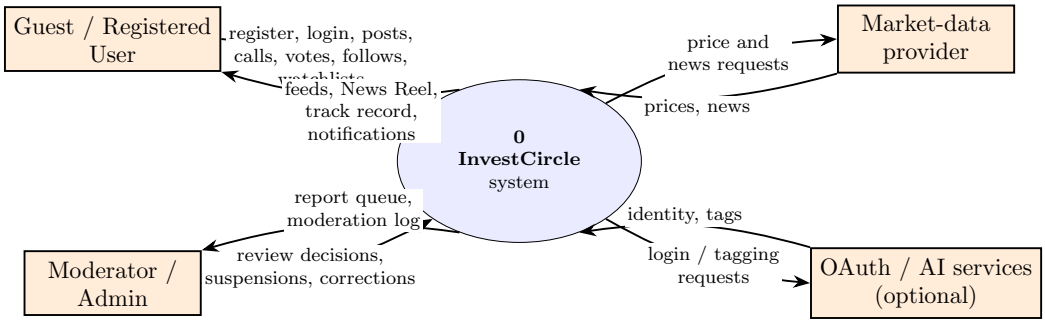


Figure 2.3a: DFD Level 0 – context diagram of InvestCircle

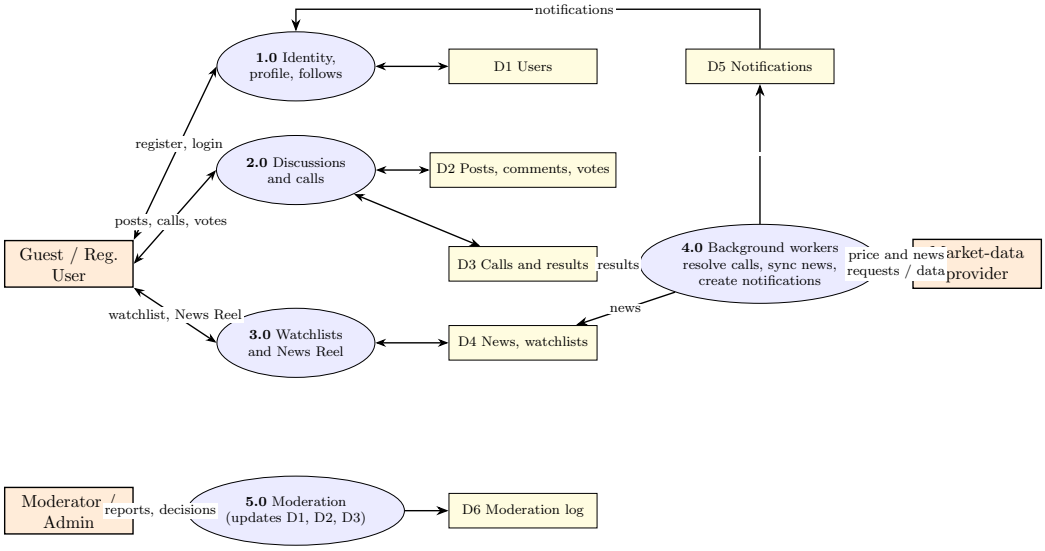


Figure 2.3b: DFD Level 1 – main processes and data stores

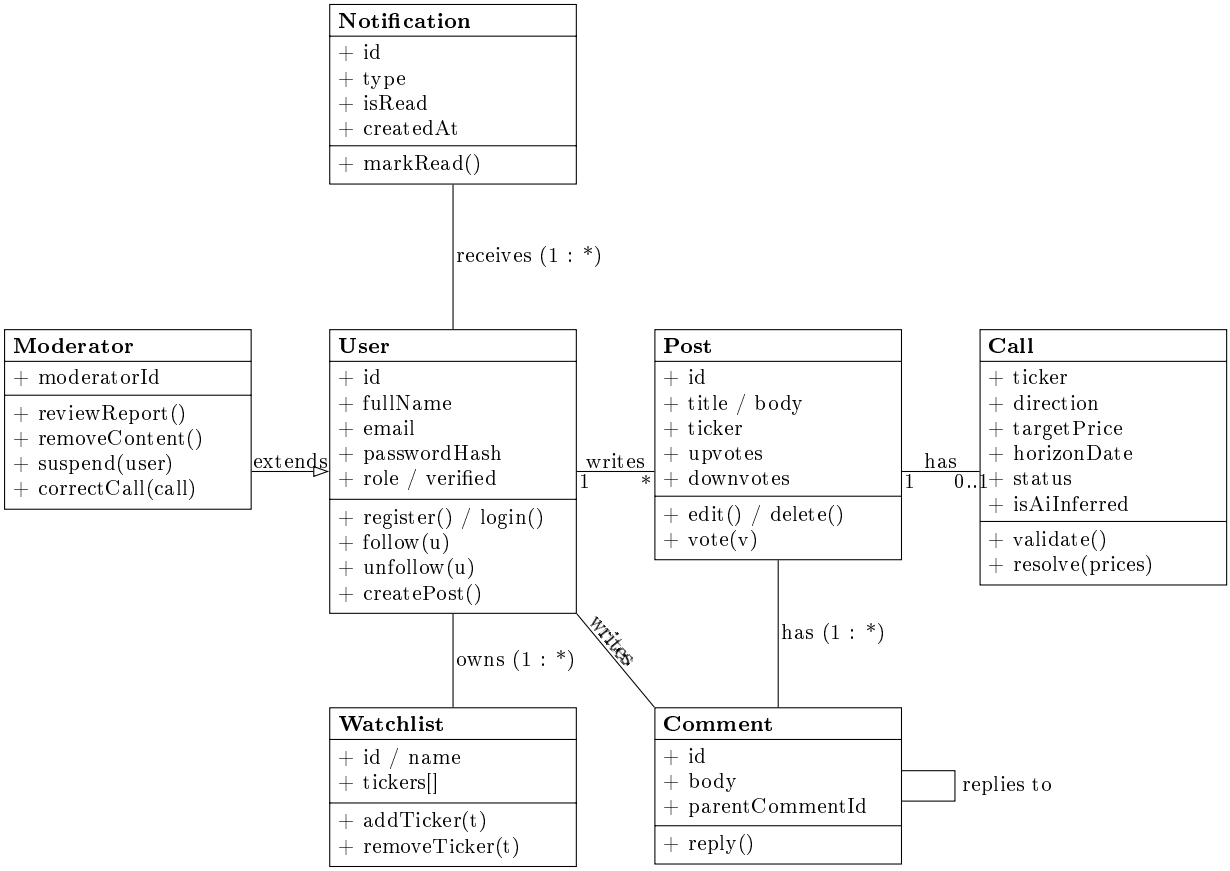


Figure 2.3c: Class diagram – domain classes

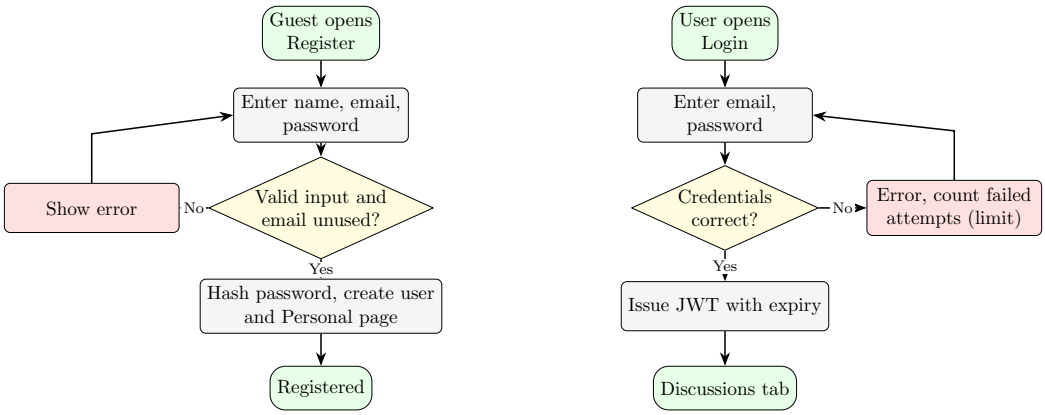


Figure 2.3d: Activity diagram – registration and login flow

3 System Features

"InvestCircle" is a stock research and call-tracking web software. So the main art of this product is to publish structured calls and resolve them automatically against real market data.

3.1 Description and Priority

"InvestCircle" has features that are main and also some are sub. The features with priority up to down -

1. User and Identity System (Must, MVP) : registration, authentication and profiles under real identity.
2. Discussion Posts and Voting (Must, MVP) : the Reddit-style Discussions tab.
3. Structured Stock Calls (Must, MVP) : this is the goal feature of this software, attached to posts and stored as Pending.
4. Follow / Unfollow and Basic Watchlists (Must, MVP).
5. Call Resolution and Track Record (Must/Should, next iteration) : the call data model is built in the MVP; automatic price-based resolution follows as soon as the core API is stable.
6. Comments and Threaded Replies, Personalised News Reel, Notifications, Search, Contributor Statistics, Basic Moderation (Should, subsequent iterations).
7. Sector/Sentiment Tagging, AI-Inferred Calls, Video Reactions, Advanced Recommendations (Could, optional extensions).

The first meaningful milestone is a working platform where a user can register, publish a structured call, follow another user, maintain a watchlist and eventually view the resolved outcome of the call.

3.2 Functional Requirements

The "InvestCircle" website is being build on React (Frontend), Node.js with a REST framework (Backend), PostgreSQL (Database), Redis (cache and jobs) and JWT authentication.

ID	Requirement	Pri.
User and identity		
FR-AUTH-01	A guest shall be able to register with email and password; OAuth login is optional.	Must
FR-AUTH-02	Passwords shall be stored only as bcrypt/Argon2 hashes; requests with missing, invalid or expired JWT shall be rejected.	Must
FR-AUTH-03	Email format and password strength shall be validated; duplicate emails rejected; users can log out.	Must
FR-PROF-01	Each user shall have a persistent Personal page with bio, follower/following counts, own posts and call history.	Must
FR-PROF-02	Users may edit only their own profile; all content shall show the author's real identity (no anonymous handles).	Must
FR-PROF-03	The Personal page shall show the accuracy score and each resolved call with its outcome.	Must
Discussions		
FR-DISC-01	A user shall be able to create a post with research text and an optional ticker.	Must
FR-DISC-02	Posts shall be shown in a paginated feed, filterable by ticker.	Must
FR-DISC-03	A user shall upvote or downvote a post (one vote per user per post, changeable).	Must
FR-DISC-04	A user may edit or delete only own posts; deleting a post shall not remove its call from the track record.	Must
FR-DISC-05	Users shall be able to comment and reply to comments, forming threads via a parent-comment relation.	Should
FR-DISC-06	Users shall be able to report posts and comments.	Should
Structured calls		
FR-CALL-01	A post may carry at most one call (ticker, direction, target price, horizon).	Must
FR-CALL-02	The system shall validate that the ticker exists, target price > 0 and horizon is a future date within an allowed maximum.	Must

ID	Requirement	Pri.
FR-CALL-03	On creation the call shall be stored with status Pending, creation time and reference price.	Must
FR-CALL-04	Core call fields shall not be editable after creation.	Must
FR-CALL-05	Calls shall carry a flag distinguishing explicit from AI-inferred calls; status shall be shown as a badge (Pending/Hit/Miss).	Must
Follow and watchlists		
FR-FOL-01	A user shall follow/unfollow another user (not self, not twice); counts and lists are visible on the Personal page.	Must
FR-WL-01	A user shall create watchlists and add/remove tickers; duplicate tickers in a list are rejected.	Must
FR-WL-02	Watchlist tickers shall serve as the interest signal for the News Reel.	Should
Call resolution and track record		
FR-RES-01	A background process shall periodically select Pending calls whose horizon has expired.	Must
FR-RES-02	For each due call the system shall fetch price data via the provider layer (cache first) and apply the Hit/Miss rule of Section 2.4.	Must
FR-RES-03	Resolution timestamp and data source shall be stored for every resolved call.	Must
FR-RES-04	If data is unavailable the call shall stay Pending and be retried with back-off; persistent failures are logged and flagged.	Must
FR-RES-05	After resolution the author's statistics (hits, misses, accuracy) shall be updated; displayed history shall match stored resolution records.	Must
FR-RES-06	Resolved calls are immutable except for logged administrator corrections.	Should
FR-RES-07	Explicit and AI-inferred calls shall be counted and shown separately, never silently mixed.	Must
News Reel, notifications, search, moderation		
FR-NEWS-01	The system shall periodically fetch market news (headline, source, timestamp, ticker, category) from the provider.	Should
FR-NEWS-02	The News Reel shall be a continuously vertically scrolling feed with cursor-based loading.	Should

ID	Requirement	Pri.
FR-NEWS-03	News matching a user's watchlist shall be boosted by a simple deterministic ranking rule (no complex recommender initially).	Should
FR-NOT-01	Notifications shall be generated for new followers, comments/replies and call resolutions, and can be marked read.	Should
FR-SRCH-01	Search shall cover users, stocks and discussions.	Should
FR-STAT-01	Contributor statistics (total calls, hit rate, per-ticker accuracy) shall be shown.	Should
FR-MOD-01	Moderators shall review reports, remove content and suspend accounts; all actions are logged.	Should
Optional extensions		
FR-AI-01	AI may classify posts by sector and sentiment (positive/neutral/negative) at creation.	Could
FR-AI-02	AI may detect implied calls in free text; they are evaluated like explicit calls and always labelled "AI-inferred".	Could
FR-AI-03	AI components shall stay outside the critical path; their failure shall not block core features.	Must
FR-VID-01	Users may record short video reactions to a news item; up to three reactions from others are shown, prioritised by verified status and follower count.	Could

4 Other Nonfunctional Requirements

4.1 Performance Requirements

"InvestCircle" will be used for peer-to-peer stock research with growing users, posts and market data. So for more interaction Node.js, React, PostgreSQL and Redis are used.

- NFR-PERF-01: discussion and news feeds shall load within approximately 1.5 seconds under the seeded project-scale workload.
- NFR-PERF-02: large feeds and profile endpoints shall use pagination or cursor-based retrieval.
- NFR-PERF-03: database queries shall remain acceptably fast as seeded data grows (indexes on user IDs, tickers, timestamps).
- NFR-PERF-04: market data and hot feed data shall be cached in Redis.
- NFR-PERF-05: market sync, call resolution, notifications and news updates shall run asynchronously in background workers.
- **Scalability:** the backend is stateless so several instances can run concurrently; data providers are isolated behind a service layer; deployment uses containers and managed PostgreSQL.

4.2 Security Requirements

No one without registered users can enter the main features of the website. One particular user of a section only can perform his/her particular actions.

- NFR-SEC-01: passwords hashed with bcrypt or Argon2, never stored or logged in plain text.
- NFR-SEC-02: token-based authentication (JWT) with expiry; all traffic over HTTPS.
- NFR-SEC-03: users can modify only resources they own; privileged operations need moderator/admin permission.
- NFR-SEC-04: server-side input validation; parameterised queries or an ORM (against SQL injection); output escaping (against XSS).
- NFR-SEC-05: rate limiting on login, posting and reporting.

- NFR-SEC-06: API keys and secrets in environment configuration, never in source control.

4.3 Software Quality Attributes

In the development phase also testing and review of the team is been continued, so that the quality of the software is been maintained and all the requirements are been fulfilled.

- **Reliability:** at least 95% of due calls resolved automatically; core features remain available when optional AI services are down; provider failures are retried and cached data is served; resolution timestamp and source are stored for audit.
- **Usability:** responsive on desktop and mobile with light and dark mode; a new user can register and publish a first post with a call in under five minutes; clear inline validation messages.
- **Maintainability:** modular layered code with explicit module ownership; Git version control; CI pipeline running automated tests on each push.
- **Testing:** unit tests (call resolution, feed ranking), integration tests (API and database), frontend tests of critical workflows and load tests of feed and profile endpoints, with at least 70% automated coverage of backend and business logic.

Acceptance criteria:

Metric	**Target**
Call-resolution reliability	$\geq 95\%$ of due calls resolved automatically
Feed performance	Feeds load in ≈ 1.5 s under seeded workload
Test coverage	$\geq 70\%$ backend and business-logic coverage
Functional completeness	All Must requirements demonstrated end-to-end
Graceful degradation	Core functionality works with AI services disabled
Data consistency	Displayed call history matches stored resolution records

Table 4.1: Acceptance criteria

4.4 Business Rules

"InvestCircle" is for peer-to-peer stock research and for tracking the outcome of structured calls.

- Discussion is identity-based; anonymous posting is not permitted.
- A post carries at most one structured call; its core fields cannot be changed after creation.
- A call's result is decided only by market data, never by the author or by votes.
- AI-inferred calls are always labelled and counted separately from confirmed calls.
- Credibility is expressed through historical accuracy, not followers or votes.
- The platform offers research and discussion only, with no trade execution.
- A user can follow another user at most once and cannot follow themselves.
- The platform shall display a disclaimer that content is community opinion, not investment advice, and that past call accuracy does not guarantee future results; market data is used per the provider's terms.

Basically increase trust and save research time.

5 Other Requirements

"InvestCircle" needs maintenance as it is a long process software. It will need re-factoring and further the requirements can be changed as the field is changing frequently. Other requirements are:

- **Documentation:** API documentation (OpenAPI), database schema documentation and a short user guide.
- **Deployment:** containerised deployment with environment-based configuration; errors, failed resolutions and moderation actions are logged. English only in the first version.
- **Phasing:** *Initial deliverable (MVP)* - registration/authentication, profiles, discussion posts, structured calls, follow/unfollow, basic watchlists, call status storage, PostgreSQL integration and REST API. *Subsequent* - automatic call resolution, News Reel, comments, notifications, search, statistics, moderation, optional tagging. *Extensions* - video reactions, AI-inferred calls, advanced recommendations and rankings. Team of four, four months.
- **Open issues to confirm:** final market-data provider and rate-limit tier; the Hit/Miss rule; maximum horizon and supported exchanges; account-deletion policy versus permanent track records.

Risks and mitigations.

- External API limits or downtime → caching, scheduled sync, provider abstraction, retry.
- Incorrect call resolution → store timestamp and data source; deterministic rule; unit tests.
- Scope creep (AI, video, recommendations) → keep optional features isolated from the MVP.
- Security risks → hashing, token auth, authorisation, validation, rate limiting.
- Performance degradation → pagination, indexing and caching from the start.
- Uneven team contribution → explicit module ownership and monthly milestones.

References. (1) InvestCircle Project Proposal, UCS503P, Thapar Institute of Engineering and Technology. (2) IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications. (3) Alpha Vantage / Finnhub API documentation.